

Maths for AIML - SEE Quiz Solutions (Part A)

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1. Fundamental Subspaces Dimensions

Question: If the null space of an 8×4 matrix A is 3-dimensional, give the dimensions of the other three fundamental subspaces.

Solution:

- **Matrix Dimensions:** $m = 8$ (rows), $n = 4$ (columns).
- **Given:** Dimension of Null Space (Nullity) = 3.
- **Rank-Nullity Theorem:** $Rank + Nullity = n$ (number of columns).
 - $r + 3 = 4 \implies r = 1$.
- **Dimensions of the Subspaces:**
 1. **Column Space ($C(A)$):** Dimension = $r = 1$.
 2. **Row Space ($C(A^T)$):** Dimension = $r = 1$.
 3. **Left Null Space ($N(A^T)$):** Dimension = $m - r = 8 - 1 = 7$.

2. Vector Space of Symmetric Matrices

Question: Select or reject with justification: "The set of all 2×2 symmetric matrices is a vector space."

Answer: Select (True)

Justification: A set is a vector space if it is closed under addition and scalar multiplication. Let S be the set of 2×2 symmetric matrices ($A^T = A$).

1. **Closure under addition:** If A and B are symmetric, then $(A + B)^T = A^T + B^T = A + B$. Thus, $A + B$ is symmetric.
2. **Closure under scalar multiplication:** If A is symmetric and c is a scalar, then $(cA)^T = c(A^T) = cA$. Thus, cA is symmetric. Since the set contains the zero matrix and is closed under linear combinations, it is a valid subspace of the vector space of all 2×2 matrices ($M_{2 \times 2}$).

3. Euclidean Norm Calculation

Question: Given the Euclidean norm, $\|u\| = 4$, $\|v\| = 3$. If u and v are perpendicular, determine $\|u + v\|$.

Solution: Using the expansion of the squared norm:

$$\|u + v\|^2 = (u + v) \cdot (u + v) = \|u\|^2 + 2(u \cdot v) + \|v\|^2$$

Since u and v are perpendicular (orthogonal), their dot product $u \cdot v = 0$.

$$\|u + v\|^2 = 4^2 + 0 + 3^2 = 16 + 9 = 25$$

$$\|u + v\| = \sqrt{25} = 5$$

4. Constructing a Symmetric Matrix

Question: Determine a real symmetric matrix whose characteristic equation is $\lambda(\lambda^2 - 4)(\lambda - 1) = 0$.

Solution: First, find the eigenvalues (roots of the characteristic equation):

1. $\lambda = 0$
2. $\lambda^2 - 4 = 0 \implies \lambda = 2, -2$
3. $\lambda - 1 = 0 \implies \lambda = 1$ Eigenvalues: $\{0, 2, -2, 1\}$.

A diagonal matrix is always symmetric. We can construct a 4×4 diagonal matrix with these eigenvalues on the main diagonal:

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

5. Heteroscedasticity

Question: Define heteroscedasticity.

Definition: In statistics and regression analysis, **heteroscedasticity** refers to the condition where the variance of the errors (residuals) is not constant across all levels of the independent variable(s). Visually, this often looks like a "cone" shape in a residual plot, where the spread of data points widens or narrows as the predicted value increases.

6. OLS vs. Weighted Least Squares

Question: How is the ordinary least squares (OLS) problem different from a weighted least squares (WLS) case?

Answer:

- **Assumption:** OLS assumes **homoscedasticity** (constant variance of errors), whereas WLS is designed to handle **heteroscedasticity** (non-constant variance).
- **Objective Function:**
 - **OLS** minimizes the sum of squared residuals: $\sum (y_i - \hat{y}_i)^2$.
 - **WLS** minimizes the *weighted* sum of squared residuals: $\sum w_i (y_i - \hat{y}_i)^2$.
 - In WLS, observations with higher variance (more noise) are given smaller weights (w_i) to reduce their influence on the model fit.

7. Acceleration Vector

Question: The position vector of point P at time t is $(3 \cos t, 5 \sin t, 4 \cos t)$. What is the acceleration?

Solution: Let position $r(t) = \langle 3 \cos t, 5 \sin t, 4 \cos t \rangle$.

1. **Velocity** ($v(t) = r'(t)$):

$$v(t) = \langle -3 \sin t, 5 \cos t, -4 \sin t \rangle$$

2. **Acceleration** ($a(t) = v'(t)$):

$$a(t) = \langle -3 \cos t, -5 \sin t, -4 \cos t \rangle$$

Note: In this specific case, $a(t) = -r(t)$.

8. Gradient Calculation

Question: Find the gradient of the function $f(x, y) = \sin(x + y)$.

Solution: The gradient ∇f is a vector of partial derivatives: $\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$.

1. $\frac{\partial f}{\partial x} = \cos(x + y) \cdot \frac{d}{dx}(x + y) = \cos(x + y)$
2. $\frac{\partial f}{\partial y} = \cos(x + y) \cdot \frac{d}{dy}(x + y) = \cos(x + y)$

$$\nabla f = \begin{bmatrix} \cos(x + y) \\ \cos(x + y) \end{bmatrix}$$

9. Pseudo Inverse of Rank-2 Matrix

Question: Consider a 2×3 real matrix A with rank of A equal to 2. What is the pseudo inverse of A ?

Solution:

- Matrix size: $m = 2, n = 3$.
- Rank $r = 2$. Since $r = m$ (the number of rows), the matrix A has **full row rank**. For a full row rank matrix, the Moore-Penrose pseudo inverse (A^+) is the **right inverse**, given by the formula:

$$A^+ = A^T(AA^T)^{-1}$$

10. Gradient Descent Issue

Question: After minimizing a function using gradient descent, if you notice that the function values are not decreasing in consecutive steps, what could be an issue?

Answer: The most likely issue is that the **learning rate (step size, α) is too large**. If the step size is too big, the algorithm can overshoot the minimum, causing the loss function to bounce back and forth across the valley, potentially leading to divergence (function values increasing) or oscillation rather than convergence.

Maths for AIML - Quiz Part 2

2.a. Fundamental Subspaces of a 3×2 Matrix

Problem: Construct a 3×2 matrix with a column space that equals the XY plane. Find the other three subspaces associated with it and mention one basis for each.

1. Constructing the Matrix (A)

- **Requirement:** The Column Space ($C(A)$) must be the XY plane in $\mathbb{R}^3$.
- **XY Plane:** The set of all vectors $\begin{bmatrix} x \\ y \\ 0 \end{bmatrix}$. This subspace has dimension 2.
- **Matrix Dimensions:** Since the column space is in $\mathbb{R}^3$, the matrix must have 3 rows. Since it is a " 3×2 " matrix (implied by "3 into 2" and the dimension needed to span a plane), it has 2 columns.
- **Construction:** We need 2 linearly independent columns that have 0 as the z-component. The standard basis vectors for the XY plane work best.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

2. The Four Fundamental Subspaces and Bases

- **i. Column Space ($C(A)$)**
 - **Definition:** The span of the columns of A .
 - **Description:** The XY plane in $\mathbb{R}^3$.
 - **Dimension:** $r = 2$.
 - **Basis:**

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

- **ii. Row Space ($C(A^T)$)**
 - **Definition:** The subspace of $\mathbb{R}^n$ (where $n = 2$) spanned by the rows of A .
 - **Matrix:** $A^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$.

- **Description:** The entire space $\mathbb{R}^2$.
- **Dimension:** $r = 2$.
- **Basis:**

$$\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

• **iii. Null Space ($N(A)$)**

- **Definition:** The set of vectors x in $\mathbb{R}^2$ such that $Ax = 0$.
- **Calculation:**

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \implies \begin{cases} x_1 = 0 \\ x_2 = 0 \end{cases}$$

- **Description:** Contains only the zero vector $\{\vec{0}\}$.
- **Dimension:** $n - r = 2 - 2 = 0$.
- **Basis:** Empty Set $\emptyset$ (The zero subspace has no basis).

• **iv. Left Null Space ($N(A^T)$)**

- **Definition:** The set of vectors y in $\mathbb{R}^3$ such that $A^T y = 0$. This is orthogonal to the column space (the Z-axis).
- **Calculation:**

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies \begin{cases} y_1 = 0 \\ y_2 = 0 \end{cases}$$

y_3 is a free variable.

- **Description:** The Z-axis in $\mathbb{R}^3$.
- **Dimension:** $m - r = 3 - 2 = 1$.
- **Basis:**

$$\left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

2.b. Statement Justification

Statement 1: "For a given $n \times n$ real matrix A , two distinct eigenvectors, one is not the scalar multiplication of the other, can have the same eigen[value]." (Note: Assuming the prompt meant "eigenvalue" instead of "eigen vectors", as distinct vectors having the same vector is impossible).

- **Answer: Select (True)**

- **Justification:** Two linearly independent eigenvectors (where one is not a scalar multiple of the other) can correspond to the same eigenvalue if the **geometric multiplicity** of that eigenvalue is greater than 1.

- **Example:** In the Identity matrix $I_{2 \times 2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, the vectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are distinct and linearly independent, but both have the eigenvalue $\lambda = 1$.

Statement 2: " $\mathbb{R}^2$ is isomorphic to the XY plane of $\mathbb{R}^3$."

- **Answer: Select (True)**

- **Justification:** Two vector spaces are isomorphic if and only if they have the **same dimension** and are over the same field.

- $\mathbb{R}^2$ has dimension 2.
- The XY plane of $\mathbb{R}^3$ (subspace of vectors with $z = 0$) has dimension 2.
- Since both are 2-dimensional real vector spaces, there exists a bijective linear transformation (isomorphism) between them.
- **Mapping:** $T : \mathbb{R}^2 \rightarrow \text{XY-plane}$ defined by $T(x, y) = (x, y, 0)$.

Maths for AIML - Quiz Part 3

3.a. Orthogonality and Linear Combinations

Given Vectors:

$$u_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad u_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \quad u_3 = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}$$

i) Are the vectors u_1, u_2, u_3 orthogonal?

To check orthogonality, we calculate the dot product of every pair. A set of vectors is mutually orthogonal if $u_i \cdot u_j = 0$ for all $i \neq j$.

1. $u_1 \cdot u_2$:

$$(1)(1) + (1)(-1) + (0)(1) = 1 - 1 + 0 = 0$$

2. $u_1 \cdot u_3$:

$$(1)(1) + (1)(-1) + (0)(-2) = 1 - 1 + 0 = 0$$

3. $u_2 \cdot u_3$:

$$(1)(1) + (-1)(-1) + (1)(-2) = 1 + 1 - 2 = 0$$

Answer: Since all pairwise dot products are zero, **YES**, the vectors are orthogonal.

ii) Are the vectors u_1, u_2, u_3 orthonormal?

A set of vectors is orthonormal if they are orthogonal AND each vector has a length (norm) of 1 (unit vectors).

Calculate the squared norms ($\|u\|^2 = u \cdot u$):

$$1. \quad \|u_1\|^2 = 1^2 + 1^2 + 0^2 = 2 \implies \|u_1\| = \sqrt{2}$$

$$2. \quad \|u_2\|^2 = 1^2 + (-1)^2 + 1^2 = 3 \implies \|u_2\| = \sqrt{3}$$

$$3. \quad \|u_3\|^2 = 1^2 + (-1)^2 + (-2)^2 = 6 \implies \|u_3\| = \sqrt{6}$$

Answer: Since the norms are not equal to 1, **NO**, they are not orthonormal.

iii) Express the vector $x = [1, 2, 3]^T$ as a linear combination of u_1, u_2, u_3 .

We want to find scalars c_1, c_2, c_3 such that:

$$x = c_1 u_1 + c_2 u_2 + c_3 u_3$$

Since the basis $\{u_1, u_2, u_3\}$ is **orthogonal**, we can use the formula:

$$c_i = \frac{x \cdot u_i}{\|u_i\|^2}$$

- **Calculate c_1 :**

$$x \cdot u_1 = (1)(1) + (2)(1) + (3)(0) = 3$$

$$\|u_1\|^2 = 2$$

$$\Rightarrow c_1 = \frac{3}{2}$$

- **Calculate c_2 :**

$$x \cdot u_2 = (1)(1) + (2)(-1) + (3)(1) = 1 - 2 + 3 = 2$$

$$\|u_2\|^2 = 3$$

$$\Rightarrow c_2 = \frac{2}{3}$$

- **Calculate c_3 :**

$$x \cdot u_3 = (1)(1) + (2)(-1) + (3)(-2) = 1 - 2 - 6 = -7$$

$$\|u_3\|^2 = 6$$

$$\Rightarrow c_3 = -\frac{7}{6}$$

Final Expression:

$$x = \frac{3}{2}u_1 + \frac{2}{3}u_2 - \frac{7}{6}u_3$$

3.b. Projection Matrices

i) **P is a projection matrix. Prove that $P^2 = P$.**

Geometric Proof: A projection matrix P maps any vector b onto a specific subspace S . Let $p = Pb$. By definition, p lies in the subspace S . If we apply the projection matrix again to p , we are projecting a vector that is *already* in the subspace onto that same subspace. Therefore, the vector should not change.

$$P(p) = p$$

Substituting $p = Pb$:

$$P(Pb) = Pb \implies P^2b = Pb$$

Since this holds for any vector b , it implies $P^2 = P$.

Algebraic Proof (using the formula): For a subspace with basis columns in matrix A , the projection matrix is $P = A(A^T A)^{-1} A^T$.

$$\begin{aligned} P^2 &= (A(A^T A)^{-1} A^T)(A(A^T A)^{-1} A^T) \\ &= A(A^T A)^{-1} (A^T A) (A^T A)^{-1} A^T \\ &\quad \underbrace{(A^T A)(A^T A)^{-1}}_I \\ &= A(A^T A)^{-1} I A^T \\ &= A(A^T A)^{-1} A^T \\ &= P \end{aligned}$$

Thus, P is idempotent.

ii) **Obtain the matrix that projects every vector in $\mathbb{R}^3$ along the line passing through the origin along the vector $a = [1, -1, 1]^T$.**

For projection onto a 1D line spanned by vector a , the projection matrix formula is:

$$P = \frac{aa^T}{a^T a}$$

- **Step 1: Calculate the denominator ($a^T a$ or $\|a\|^2$)**

$$a^T a = \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 1^2 + (-1)^2 + 1^2 = 3$$

- **Step 2: Calculate the numerator (aa^T)** This results in a 3×3 matrix.

$$aa^T = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} (1)(1) & (1)(-1) & (1)(1) \\ (-1)(1) & (-1)(-1) & (-1)(1) \\ (1)(1) & (1)(-1) & (1)(1) \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

- **Step 3: Combine to find P**

$$P = \frac{1}{3} \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

Maths for AIML - Quiz Part 4

4.a. Orthogonal Complement Subspace

Problem: Obtain the orthogonal complement subspace of the line passing through the origin along the vector $v = [1, -1, 1]^T$.

Solution: The orthogonal complement of a line (1-dimensional subspace) in $\mathbb{R}^3$ is a plane (2-dimensional subspace) passing through the origin.

Let W be the subspace spanned by $v = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$. The orthogonal complement $W^\perp$ consists of

all vectors $x = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ such that $x \cdot v = 0$.

1. Equation of the Subspace:

$$x \cdot v = 0$$

$$(x)(1) + (y)(-1) + (z)(1) = 0$$

$$x - y + z = 0$$

This is the Cartesian equation of the plane representing the orthogonal complement.

2. Basis for the Subspace: To find a basis, we solve the equation $x - y + z = 0$ for the pivot variable (say, x) in terms of free variables (y and z).

$$x = y - z$$

Express the general vector:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} y - z \\ y \\ z \end{bmatrix} = y \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Thus, a basis for the orthogonal complement is:

$$\text{Basis} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

4.b. Principal Component Analysis (PCA)

Problem: Data Points: $x_1 = [1, 1]$, $x_2 = [1, 1]$, $x_3 = [2, 0]$, $x_4 = [0, 2]$. Tasks:

1. Compute unit length principal component directions.
2. State which one PCA would choose for 1D representation.
3. Obtain projections of all 4 sample points onto the principal direction.

Step 1: Center the Data (Compute Mean)

$$\mu = \frac{1}{4} \sum x_i = \frac{1}{4}([1, 1] + [1, 1] + [2, 0] + [0, 2]) = \frac{1}{4}[4, 4] = [1, 1]^T$$

Subtract the mean from each data point to get centered data ($\hat{x}_i = x_i - \mu$):

- $\hat{x}_1 = [1 - 1, 1 - 1]^T = [0, 0]^T$
- $\hat{x}_2 = [1 - 1, 1 - 1]^T = [0, 0]^T$
- $\hat{x}_3 = [2 - 1, 0 - 1]^T = [1, -1]^T$
- $\hat{x}_4 = [0 - 1, 2 - 1]^T = [-1, 1]^T$

Step 2: Compute Covariance Matrix (C) For samples, $C = \frac{1}{N-1} \sum \hat{x}_i \hat{x}_i^T$ (where $N = 4$).

$$\sum \hat{x}_i \hat{x}_i^T = \begin{bmatrix} 0 \\ 0 \end{bmatrix} [0 \ 0] + \begin{bmatrix} 0 \\ 0 \end{bmatrix} [0 \ 0] + \begin{bmatrix} 1 \\ -1 \end{bmatrix} [1 \ -1] + \begin{bmatrix} -1 \\ 1 \end{bmatrix} [-1 \ 1]$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$$

$$C = \frac{1}{3} \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} = \begin{bmatrix} 2/3 & -2/3 \\ -2/3 & 2/3 \end{bmatrix}$$

Step 3: Eigenvalues and Eigenvectors Find roots of $|\lambda I - C| = 0$:

$$\left(\frac{2}{3} - \lambda\right)^2 - \left(-\frac{2}{3}\right)^2 = 0$$

$$\lambda^2 - \frac{4}{3}\lambda + \frac{4}{9} - \frac{4}{9} = 0$$

$$\lambda(\lambda - \frac{4}{3}) = 0$$

Eigenvalues: $\lambda_1 = \frac{4}{3} \approx 1.33$ and $\lambda_2 = 0$.

Eigenvectors:

1. For $\lambda_1 = 4/3$: $(\frac{2}{3} - \frac{4}{3})x - \frac{2}{3}y = 0 \implies -\frac{2}{3}x - \frac{2}{3}y = 0 \implies x = -y$. Vector: $[1, -1]^T$. **Unit Vector (u_1):** $\begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$

2. For $\lambda_2 = 0$: $\frac{2}{3}x - \frac{2}{3}y = 0 \implies x = y$. Vector: $[1, 1]^T$. **Unit Vector (u_2):** $\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$

Step 4: PCA Choice PCA chooses the direction corresponding to the **largest eigenvalue** (highest variance).

- $\lambda_1 = 4/3 > \lambda_2 = 0$.
- **Chosen Principal Component:** $u_1 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} \approx \begin{bmatrix} 0.707 \\ -0.707 \end{bmatrix}$.

Step 5: Projections (1D Representation) The projection of the data points onto the principal component is calculated by the dot product of the **centered** data with u_1 .

$$z_i = \hat{x}_i \cdot u_1$$

1. **Point 1 (1, 1):** $\hat{x}_1 = [0, 0]$.

$$z_1 = 0 \cdot \frac{1}{\sqrt{2}} + 0 \cdot \frac{-1}{\sqrt{2}} = 0$$

2. **Point 2 (1, 1):** $\hat{x}_2 = [0, 0]$.

$$z_2 = 0 \cdot \frac{1}{\sqrt{2}} + 0 \cdot \frac{-1}{\sqrt{2}} = 0$$

3. **Point 3 (2, 0):** $\hat{x}_3 = [1, -1]$.

$$z_3 = 1(\frac{1}{\sqrt{2}}) + (-1)(\frac{-1}{\sqrt{2}}) = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2} (\approx 1.414)$$

4. **Point 4 (0, 2):** $\hat{x}_4 = [-1, 1]$.

1 -1 1 1 2 -

Maths for AIML - Quiz Part 5

5.a. Statistical Estimation

Problem: Suppose you measure your pulse rate x 3 times and get values $x = B_1, x = B_2, x = B_3$. If each measurement is independent with zero mean (errors) and unit variances, find the best estimate $\hat{x}$ based on B_1, B_2, B_3 . What is the corresponding variance?

Solution:

We can model this problem using the linear model $b = Ax + e$, where we try to estimate the scalar x .

- **Measurements:** $b = \begin{bmatrix} B_1 \\ B_2 \\ B_3 \end{bmatrix}$
- **Design Matrix:** $A = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ (Since every measurement observes x directly).
- **Noise/Error:** The errors are independent with variance $\sigma^2 = 1$.

1. Best Estimate ($\hat{x}$) For a system $Ax = b$, the best estimate (Least Squares or Maximum Likelihood Estimate under Gaussian assumptions) is given by the normal equation:

$$\hat{x} = (A^T A)^{-1} A^T b$$

- **Step 1: Calculate $A^T A$**

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 1 + 1 + 1 = 3$$

- **Step 2: Calculate $A^T b$**

$$A^T b = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \\ B_3 \end{bmatrix} = B_1 + B_2 + B_3$$

- **Step 3: Solve for $\hat{x}$**

$$\hat{x} = (3)^{-1}(B_1 + B_2 + B_3) = \frac{B_1 + B_2 + B_3}{3}$$

Result: The best estimate is simply the **sample mean** (average) of the measurements.

2. Variance of the Estimate The variance of the Least Squares estimator is given by the formula:

$$\text{Var}(\hat{x}) = \sigma^2(A^T A)^{-1}$$

- Given unit variance, $\sigma^2 = 1$.
- We calculated $(A^T A) = 3$, so $(A^T A)^{-1} = \frac{1}{3}$.

$$\text{Var}(\hat{x}) = 1 \cdot \frac{1}{3} = \frac{1}{3}$$

Alternative Derivation (using scalar variance properties):

$$\text{Var}(\hat{x}) = \text{Var}\left(\frac{B_1 + B_2 + B_3}{3}\right)$$

Since measurements are independent:

$$= \frac{1}{3^2}[\text{Var}(B_1) + \text{Var}(B_2) + \text{Var}(B_3)]$$

$$= \frac{1}{9}[1 + 1 + 1] = \frac{3}{9} = \frac{1}{3}$$

Answer:

- **Best Estimate:** $\hat{x} = \frac{B_1 + B_2 + B_3}{3}$
- **Variance:** $\frac{1}{3}$

5.b. Gaussian Mixture Models (GMM)

Problem: List out three limitations of Gaussian Mixture Models.

Answer:

1. **Non-Convex Optimization (Local Minima):** GMMs are typically trained using the Expectation-Maximization (EM) algorithm. The likelihood function for a GMM is not convex, meaning the algorithm can get stuck in **local optima**. The final result is highly sensitive to the random initialization of the parameters.

2. **Number of Components (k) is Unknown:** Like K-Means, standard GMMs require the user to specify the number of Gaussian components (k) beforehand. If k is incorrect, the model will either underfit (merging distinct clusters) or overfit (splitting a single cluster into multiple Gaussians) the data.
3. **Assumption of Gaussian Shape:** GMMs assume that each cluster can be modeled by a Gaussian distribution (elliptical shapes). They struggle to model data with **complex, non-elliptical geometries** (e.g., concentric circles, spirals, or irregular manifolds) effectively compared to density-based methods like DBSCAN.

Maths for AIML - Quiz Part 6

6.a. Limitations of the Expectation-Maximization (EM) Algorithm

Problem: Let us look at four limitations of the expectation-maximization algorithm.

Answer:

1. **Convergence to Local Optima (Not Global):** The EM algorithm guarantees that the likelihood increases with every iteration, but it only guarantees convergence to a **local maximum**, not necessarily the global maximum. If the likelihood function is complex with many peaks (non-convex), the algorithm often gets stuck in a suboptimal solution.
2. **Sensitivity to Initialization:** Because of the local optima problem, the final result is highly dependent on the **initial values** of the parameters. Poor initialization can lead to poor convergence or getting stuck in a bad local maximum. It is common practice to run EM multiple times with different random initializations.
3. **Slow Convergence:** The rate of convergence can be very slow, especially when the "missing data" (latent variables) carries a lot of information or when the distributions (e.g., Gaussian components) have significant overlap. It may take many iterations to reach the final solution, making it computationally expensive for large datasets.
4. **Assumption of the Underlying Model:** EM assumes that the **parametric form** of the model (e.g., a Mixture of Gaussians) is known and correct. It does not learn the structure of the model itself. If the data does not fit the assumed distribution, or if the number of latent components (e.g., number of clusters k) is specified incorrectly, the algorithm will yield poor results.

6.b. Applications of Kalman Filters

Problem: List 4 applications of Kalman filters.

Answer:

1. **Navigation and GPS Systems:** Kalman filters are ubiquitous in GPS receivers (in phones, cars, and aircraft) to smooth out noisy satellite signals. They estimate the precise position and velocity of a vehicle by predicting the future state based on physics and correcting it with the noisy GPS measurements.
2. **Computer Vision (Object Tracking):** In video processing, Kalman filters are used to track moving objects (like cars or pedestrians) across frames. Even if an object is temporarily occluded (blocked from view) or the detection is noisy, the filter predicts the object's likely location based on its previous trajectory.

3. **Robotics (Sensor Fusion):** Robots use Kalman filters to fuse data from multiple sensors—such as accelerometers (IMU), gyroscopes, and wheel encoders—to determine their orientation and position. The filter weighs the sensors based on their known uncertainty (variance) to provide an estimate better than any single sensor could provide.
4. **Economics and Finance:** Kalman filters are used in time-series analysis for algorithmic trading and econometrics. They help in estimating hidden states, such as the "true" value of a volatile stock or time-varying parameters in a financial model, separating the signal (trend) from the noise (market volatility).

Maths for AIML - Quiz Part 7

7. Vector Calculus and Quadratic Forms

Problem Statement: General 2nd degree vector polynomial is of the form

$f(x) = x^T A x + b^T x$, where:

- x is a vector $[x_1, x_2, \dots, x_d]^T$.
- b is a vector $[b_1, b_2, \dots, b_d]^T$.
- A is a matrix with (i, j) -th element A_{ij} .
- A and b are constants.

i) Expand $f(x)$ for the two-dimensional case where $x = [x, y]^T$.

Solution: Let the dimension $d = 2$.

- Vector $x = \begin{bmatrix} x \\ y \end{bmatrix}$.
- Vector $b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$.
- Matrix $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$.

The function is given by the sum of the quadratic term ($x^T A x$) and the linear term ($b^T x$).

1. Linear Term ($b^T x$):

$$b^T x = \begin{bmatrix} b_1 & b_2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = b_1 x + b_2 y$$

2. Quadratic Term ($x^T A x$):

$$\begin{aligned} x^T A x &= \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \\ &= \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} A_{11}x + A_{12}y \\ A_{21}x + A_{22}y \end{bmatrix} \\ &= x(A_{11}x + A_{12}y) + y(A_{21}x + A_{22}y) \\ &= A_{11}x^2 + A_{12}xy + A_{21}yx + A_{22}y^2 \end{aligned}$$

(Note: scalars commute, so $yx = xy$).

Total Expansion:

$$f(x, y) = A_{11}x^2 + (A_{12} + A_{21})xy + A_{22}y^2 + b_1x + b_2y$$

ii) Which definition of the derivative do we need in order to evaluate $df(x)/dx$?

Solution: Since $f(x)$ is a **scalar-valued function** ($f : \mathbb{R}^n \rightarrow \mathbb{R}$) and x is a **vector** ($x \in \mathbb{R}^n$), the derivative we need is the **Gradient Vector** (often denoted as ∇f or $\frac{df}{dx}$).

Definition: The gradient is the column vector of first-order partial derivatives of f with respect to each component of x :

$$\frac{df(x)}{dx} = \nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_d} \end{bmatrix}$$

iii) Assume $x, b \in \mathbb{R}^2$ and $A \in \mathbb{R}^{2 \times 2}$. Evaluate $d(f(x))/dx$.

Solution: Using the expansion from Part (i):

$$f(x, y) = A_{11}x^2 + A_{12}xy + A_{21}xy + A_{22}y^2 + b_1x + b_2y$$

We compute the partial derivatives with respect to the components of vector $x = [x, y]^T$.

1. Partial with respect to x :

$$\begin{aligned} \frac{\partial f}{\partial x} &= \frac{\partial}{\partial x}(A_{11}x^2) + \frac{\partial}{\partial x}(A_{12}xy) + \frac{\partial}{\partial x}(A_{21}xy) + \frac{\partial}{\partial x}(A_{22}y^2) + \frac{\partial}{\partial x}(b_1x) + \frac{\partial}{\partial x}(b_2y) \\ &= 2A_{11}x + A_{12}y + A_{21}y + 0 + b_1 + 0 \\ &= 2A_{11}x + (A_{12} + A_{21})y + b_1 \end{aligned}$$

2. Partial with respect to y :

$$\begin{aligned} \frac{\partial f}{\partial y} &= \frac{\partial}{\partial y}(A_{11}x^2) + \frac{\partial}{\partial y}(A_{12}xy) + \frac{\partial}{\partial y}(A_{21}xy) + \frac{\partial}{\partial y}(A_{22}y^2) + \frac{\partial}{\partial y}(b_1x) + \frac{\partial}{\partial y}(b_2y) \\ &= 0 + A_{12}x + A_{21}x + 2A_{22}y + 0 + b_2 \\ &= (A_{12} + A_{21})x + 2A_{22}y + b_2 \end{aligned}$$

3. Assemble the Gradient Vector:

$$\frac{df(x)}{dx} = \begin{bmatrix} 2A_{11}x + (A_{12} + A_{21})y + b_1 \\ (A_{12} + A_{21})x + 2A_{22}y + b_2 \end{bmatrix}$$

We can decompose this vector back into matrix form to see the structure:

$$\begin{bmatrix} 2A_{11} & A_{12} + A_{21} \\ A_{12} + A_{21} & 2A_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

This is equivalent to $(A + A^T)x + b$.

iv) Generalize the previous result to when $x, b \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$. Evaluate $d(f(x))/dx$ and express in matrix form.

Solution: Let $f(x) = \sum_{i=1}^n \sum_{j=1}^n A_{ij}x_i x_j + \sum_{k=1}^n b_k x_k$.

We want to find the k -th component of the gradient vector, which is $\frac{\partial f}{\partial x_k}$.

1. Derivative of the Linear Term:

$$\frac{\partial}{\partial x_k} \left(\sum_{m=1}^n b_m x_m \right) = b_k$$

2. Derivative of the Quadratic Term: Using the product rule on the double sum

$\sum_i \sum_j A_{ij}x_i x_j$: The term x_k appears when $i = k$ and when $j = k$.

$$\begin{aligned} \frac{\partial}{\partial x_k} \left(\sum_{i,j} A_{ij}x_i x_j \right) &= \sum_j A_{kj}x_j + \sum_i A_{ik}x_i \\ &= (k\text{-th row of } A) \cdot x + (k\text{-th col of } A) \cdot x \end{aligned}$$

- The first sum $\sum_j A_{kj}x_j$ is the k -th element of the vector Ax .
- The second sum $\sum_i A_{ik}x_i$ is the k -th element of the vector $A^T x$ (since $A_{ik} = (A^T)_{ki}$).

3. Total Gradient: The gradient vector is formed by stacking these k components:

$$\nabla f(x) = Ax + A^T x + b$$

Matrix Form:

$$\frac{df(x)}{dx} = (A + A^T)x + b$$

v) What happens when A is a symmetric matrix ($A^T = A$)?

Solution: If A is symmetric, then the transpose of A is equal to A itself ($A^T = A$).

Substituting this into the general result from Part (iv):

$$\begin{aligned}\frac{df(x)}{dx} &= (A + A)x + b \\ &= 2Ax + b\end{aligned}$$

Conclusion: For a symmetric matrix A , the gradient of the quadratic form $x^T Ax$ is simply $2Ax$.

This is analogous to the scalar calculus rule where $\frac{d}{dx}(ax^2) = 2ax$.

Maths for AIML - Quiz Part 8

8. Derivative of a Composite Function

Problem Statement:

- Input vector: x is a 3×1 vector ($x \in \mathbb{R}^3$).
- Inner Function: $f(x) = \begin{bmatrix} x^T A x \\ b^T x \end{bmatrix}$. This is a 2×1 vector.
 - A is a 3×3 matrix.
 - b is a 3×1 vector.
- Outer Function: $g(z) = z^T z$. This is a scalar (1×1) function of a 2×1 vector z .
- Composite Function: $y(x) = g(f(x))$.
- **Task:** Compute $\frac{dy(x)}{dx}$ (The gradient of y with respect to x).

Step 1: Analyze the Outer Function $g(z)$

Let the intermediate variable be $z = f(x) = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$. The function $g(z)$ is the squared Euclidean norm:

$$g(z) = z^T z = \begin{bmatrix} z_1 & z_2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = z_1^2 + z_2^2$$

We compute the gradient of g with respect to z ($\nabla_z g$):

$$\frac{dg}{dz} = \nabla g(z) = \begin{bmatrix} \frac{\partial g}{\partial z_1} \\ \frac{\partial g}{\partial z_2} \end{bmatrix} = \begin{bmatrix} 2z_1 \\ 2z_2 \end{bmatrix} = 2z$$

Step 2: Analyze the Inner Function $f(x)$ and its Jacobian

We need the **Jacobian Matrix** of $f(x)$, denoted as J_f or $\frac{df}{dx}$. Since $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, the Jacobian will be a 2×3 matrix containing the gradients of each component row-wise.

$$f(x) = \begin{bmatrix} f_1(x) \\ f_2(x) \end{bmatrix} = \begin{bmatrix} x^T A x \\ b^T x \end{bmatrix}$$

1. Gradient of the Quadratic Component $f_1(x) = x^T A x$: Using standard vector calculus results (where $\frac{d}{dx}(x^T A x) = (A + A^T)x$):

$$\nabla f_1(x) = (A + A^T)x$$

This forms the first row of the Jacobian (transposed).

2. Gradient of the Linear Component $f_2(x) = b^T x$: Using standard vector calculus results (where $\frac{d}{dx}(b^T x) = b$):

$$\nabla f_2(x) = b$$

This forms the second row of the Jacobian (transposed).

Constructing the Jacobian Matrix $\frac{df}{dx}$:

$$\frac{df}{dx} = \begin{bmatrix} (\nabla f_1)^T \\ (\nabla f_2)^T \end{bmatrix} = \begin{bmatrix} x^T(A + A^T) \\ b^T \end{bmatrix}$$

(Note: Since $(A + A^T)$ is symmetric, its transpose is itself).

Step 3: Apply the Chain Rule

The Chain Rule for the gradient of a composite scalar function $y(x) = g(f(x))$ is:

$$\frac{dy}{dx} = \left(\frac{df}{dx} \right)^T \frac{dg}{dz} \Big|_{z=f(x)}$$

1. Transpose of the Jacobian:

$$\left(\frac{df}{dx} \right)^T = \begin{bmatrix} x^T(A + A^T) \\ b^T \end{bmatrix}^T = \begin{bmatrix} (A + A^T)x & b \end{bmatrix}$$

This is a 3×2 matrix.

2. Gradient of g evaluated at $f(x)$:

$$\frac{dg}{dz} \Big|_{z=f(x)} = 2f(x) = 2 \begin{bmatrix} x^T A x \\ b^T x \end{bmatrix}$$

This is a 2×1 vector.

Step 4: Final Calculation

Multiply the matrices to find the gradient $\frac{dy}{dx}$:

$$\begin{aligned}\frac{dy}{dx} &= [(A + A^T)x \quad b] \left(2 \begin{bmatrix} x^T Ax \\ b^T x \end{bmatrix} \right) \\ &= 2 [((A + A^T)x) \cdot (x^T Ax) + b \cdot (b^T x)]\end{aligned}$$

To simplify the notation, let the scalar value of the quadratic form be $\alpha = x^T Ax$ and the scalar value of the linear form be $\beta = b^T x$.

$$\frac{dy}{dx} = 2\alpha(A + A^T)x + 2\beta b$$

Expanded Final Answer:

$$\frac{dy(x)}{dx} = 2(x^T Ax)(A + A^T)x + 2(b^T x)b$$

Maths for AIML - Quiz Part 9

9.a. Stationary Point Analysis

Problem: The function $f(x, y, z) = 2x^2 + 2y^2 + z^2 - x - y - 2xy$ defined on $\mathbb{R}^3$ has only one stationary point. Determine it and show that it is a local minimum. (6 marks)

Solution:

Step 1: Find the Gradient Vector ∇f To find the stationary points, we calculate the first-order partial derivatives with respect to x , y , and z and set them to zero.

$$\begin{aligned}\frac{\partial f}{\partial x} &= 4x - 1 - 2y \\ \frac{\partial f}{\partial y} &= 4y - 1 - 2x \\ \frac{\partial f}{\partial z} &= 2z\end{aligned}$$

Step 2: Solve the System of Linear Equations Set the gradient to zero:

1. $4x - 2y = 1$
2. $-2x + 4y = 1$
3. $2z = 0 \implies z = 0$

Solve for x and y using (1) and (2). From (2), multiply by 2: $-4x + 8y = 2$ Add this to equation (1): $(4x - 2y) + (-4x + 8y) = 1 + 2$ $6y = 3 \implies y = \frac{1}{2}$

Substitute $y = 1/2$ back into (1): $4x - 2(1/2) = 1$ $4x - 1 = 1 \implies 4x = 2 \implies x = \frac{1}{2}$

Stationary Point: $P = (\frac{1}{2}, \frac{1}{2}, 0)$.

Step 3: Verify it is a Local Minimum (Hessian Matrix) We compute the Hessian matrix of second-order partial derivatives (H).

$$H = \begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{bmatrix} = \begin{bmatrix} 4 & -2 & 0 \\ -2 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

To determine the nature of the stationary point, we check if H is **positive definite**. We can use Sylvester's Criterion (checking leading principal minors D_i).

1. $D_1 = |4| = 4 > 0$

$$2. D_2 = \begin{vmatrix} 4 & -2 \\ -2 & 4 \end{vmatrix} = (4)(4) - (-2)(-2) = 16 - 4 = 12 > 0$$

$$3. D_3 = \det(H) = 2 \cdot \begin{vmatrix} 4 & -2 \\ -2 & 4 \end{vmatrix} = 2(12) = 24 > 0$$

Conclusion: Since all leading principal minors are positive, the Hessian matrix is **positive definite**. Therefore, the stationary point $(\frac{1}{2}, \frac{1}{2}, 0)$ is a **Local Minimum**.

9.b. Least Squares Solution

Problem: Find the least square solution to the following system of equations:

$$1. x_1 + x_2 = 2$$

$$2. x_1 - x_2 = 0$$

$$3. 2x_1 + x_2 = 4 \text{ (10 marks)}$$

Solution:

We represent this system in the matrix form $Ax = b$.

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 1 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad b = \begin{bmatrix} 2 \\ 0 \\ 4 \end{bmatrix}$$

Since the system is overdetermined (3 equations, 2 unknowns), we solve the **Normal Equation**:

$$A^T A \hat{x} = A^T b$$

Step 1: Compute $A^T A$

$$\begin{aligned} A^T A &= \begin{bmatrix} 1 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} (1)(1) + (1)(1) + (2)(2) & (1)(1) + (1)(-1) + (2)(1) \\ (1)(1) + (-1)(1) + (1)(2) & (1)(1) + (-1)(-1) + (1)(1) \end{bmatrix} \\ &= \begin{bmatrix} 1 + 1 + 4 & 1 - 1 + 2 \\ 1 - 1 + 2 & 1 + 1 + 1 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix} \end{aligned}$$

Step 2: Compute $A^T b$

$$A^T b = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} (1)(2) + (1)(0) + (2)(4) \\ (1)(2) + (-1)(0) + (1)(4) \end{bmatrix} = \begin{bmatrix} 2 + 0 + 8 \\ 2 + 0 + 4 \end{bmatrix} = \begin{bmatrix} 10 \\ 6 \end{bmatrix}$$

Step 3: Solve for $\hat{x}$ We now solve $\begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 6 \end{bmatrix}$.

Calculate the inverse of $A^T A$:

$$\det(A^T A) = (6)(3) - (2)(2) = 18 - 4 = 14$$

$$(A^T A)^{-1} = \frac{1}{14} \begin{bmatrix} 3 & -2 \\ -2 & 6 \end{bmatrix}$$

Multiply by $A^T b$:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 3 & -2 \\ -2 & 6 \end{bmatrix} \begin{bmatrix} 10 \\ 6 \end{bmatrix}$$

$$= \frac{1}{14} \begin{bmatrix} (3)(10) + (-2)(6) \\ (-2)(10) + (6)(6) \end{bmatrix}$$

$$= \frac{1}{14} \begin{bmatrix} 30 - 12 \\ -20 + 36 \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 18 \\ 16 \end{bmatrix} = \begin{bmatrix} \frac{9}{7} \\ \frac{8}{7} \end{bmatrix}$$

Final Answer:

Maths for AIML - Quiz Part 10

10. Optimization using Lagrange Multipliers

Problem Statement: Use Lagrange multipliers to find the minimum value of the function

$f(x, y) = x^2 + y^2 - 4x - 4y$ subject to the constraint $x + y = 2$. (16 marks)

Solution:

Step 1: Define the Functions

- **Objective Function (to minimize):**

$$f(x, y) = x^2 + y^2 - 4x - 4y$$

- **Constraint Function:**

$$g(x, y) = x + y - 2 = 0$$

Step 2: Form the Lagrangian Function The Lagrangian $\mathcal{L}(x, y, \lambda)$ is defined as:

$$\mathcal{L}(x, y, \lambda) = f(x, y) - \lambda \cdot g(x, y)$$

Substituting our functions:

$$\mathcal{L}(x, y, \lambda) = (x^2 + y^2 - 4x - 4y) - \lambda(x + y - 2)$$

Step 3: Find Partial Derivatives To find the critical points, we compute the partial derivatives of $\mathcal{L}$ with respect to x , y , and λ , and set them to zero.

1. **Partial with respect to x :**

$$\frac{\partial \mathcal{L}}{\partial x} = 2x - 4 - \lambda = 0 \implies 2x = \lambda + 4 \quad \text{--- (Eq. 1)}$$

2. **Partial with respect to y :**

$$\frac{\partial \mathcal{L}}{\partial y} = 2y - 4 - \lambda = 0 \implies 2y = \lambda + 4 \quad \text{--- (Eq. 2)}$$

3. **Partial with respect to λ :**

$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(x + y - 2) = 0 \implies x + y = 2 \quad \text{--- (Eq. 3)}$$

Step 4: Solve the System of Equations We have a system of three linear equations.

- From **(Eq. 1)** and **(Eq. 2)**: Since both $2x$ and $2y$ equal $\lambda + 4$, we can equate them:

$$2x = 2y \implies x = y$$

- Substitute $x = y$ into the constraint **(Eq. 3)**:

$$x + x = 2$$

$$2x = 2$$

$$x = 1$$

- Since $y = x$, then:

$$y = 1$$

- (Optional) Find λ : Using **(Eq. 1)**: $\lambda = 2(1) - 4 = -2$.

Critical Point: $(x, y) = (1, 1)$.

Step 5: Calculate the Minimum Value Substitute the critical point $(1, 1)$ back into the original objective function $f(x, y)$.

$$f(1, 1) = (1)^2 + (1)^2 - 4(1) - 4(1)$$

$$f(1, 1) = 1 + 1 - 4 - 4$$

$$f(1, 1) = 2 - 8$$

$$f(1, 1) = -6$$

Step 6: Verification (Second Order Conditions) To confirm this is a minimum (and strictly required for full marks), we can check the **Bordered Hessian Matrix** ($\bar{H}$).

$$\bar{H} = \begin{bmatrix} 0 & g_x & g_y \\ g_x & \mathcal{L}_{xx} & \mathcal{L}_{xy} \\ g_y & \mathcal{L}_{yx} & \mathcal{L}_{yy} \end{bmatrix}$$

- Constraint derivatives: $g_x = 1, g_y = 1$.
- Lagrangian second derivatives:
 - $\mathcal{L}_{xx} = \frac{\partial}{\partial x}(2x - 4 - \lambda) = 2$
 - $\mathcal{L}_{yy} = \frac{\partial}{\partial y}(2y - 4 - \lambda) = 2$
 - $\mathcal{L}_{xy} = 0$

$$\bar{H} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

Calculate the determinant $|\bar{H}|$:

$$|\bar{H}| = 0(4 - 0) - 1(2 - 0) + 1(0 - 2)$$

$$|\bar{H}| = -2 - 2 = -4$$

Conclusion: Since the determinant of the Bordered Hessian is **negative** ($|\bar{H}| < 0$), the stationary point is a **Local Minimum**.